# FLIPKART

# Customer Support Analytics

Exploratory Data Analysis & Business Insights

A concise, decision-focused report based on the supplied Jupyter notebook and project presentation. It highlights the strongest patterns rather than reproducing every chart.

| 85,907 | 19 | 79.3% | –0.1776 |
|---|---|---|---|
| Records | Columns | Inbound share | Spearman ρ: delay vs CSAT |

## Executive summary

**Customer satisfaction is generally strong:** CSAT 5 is the most common rating, followed by CSAT 4. However, the low-score segment remains an actionable service-quality opportunity.

**Inbound support dominates demand:** it accounts for approximately 79.3% of interactions. Improvements to inbound handling are therefore likely to affect the largest share of customer contacts.

**Response delay is associated with lower satisfaction:** Spearman correlation is –0.1776 ($p < 0.001$). The relationship is statistically significant but modest in strength; it should not be interpreted as proof that delay alone causes lower CSAT.

**Performance is not uniform:** the notebook reports differences by shift and variation across channel, tenure group and manager. These patterns support targeted reviews rather than one-size-fits-all interventions.

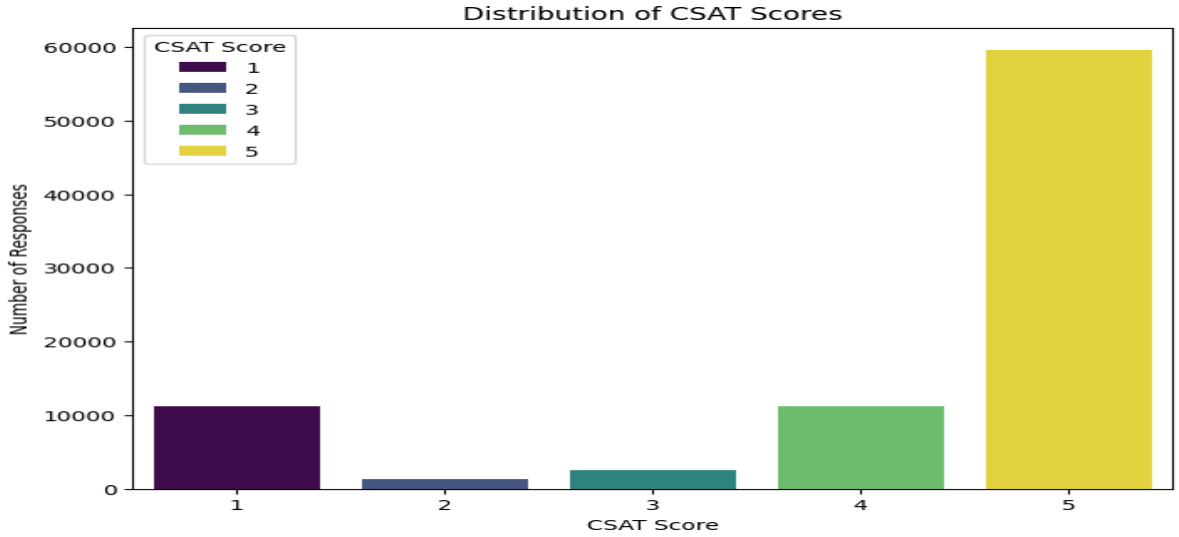
**Decision focus: reduce avoidable response delays, align capacity with inbound demand, and investigate low-CSAT cases by shift, channel and agent experience.**

Source: supplied Flipkart_Customer_support_EDA.ipynb and Flipkart project presentation. All findings below are limited to the supplied analysis.

# 1. Customer experience & channel demand

## CSAT profile

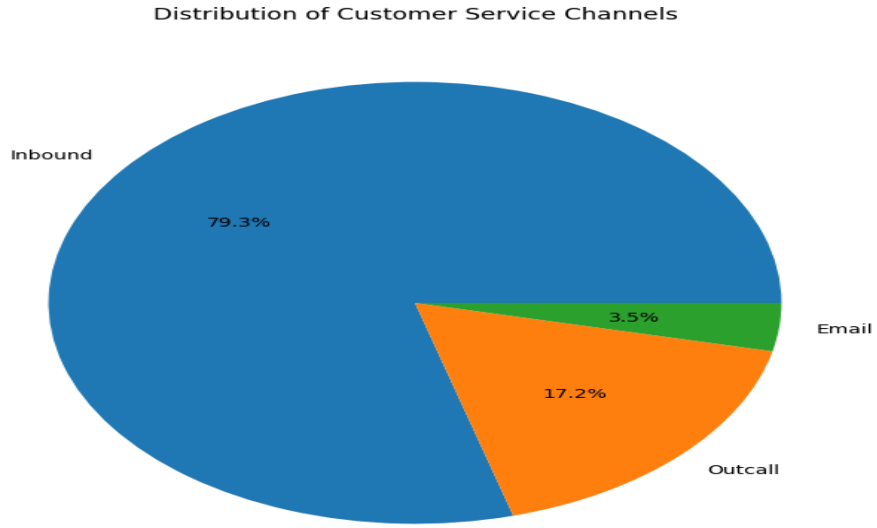
The notebook's CSAT count plot shows rating 5 as the dominant group, with rating 4 next. This is a positive overall signal, but the visible rating-1 group should be reviewed to understand recurring dissatisfaction drivers.




Business implication: protect the high-satisfaction base while using low-score cases for root-cause analysis. Segment complaints by issue type, channel and shift before choosing corrective actions.

## Customer service channel mix

Inbound represents approximately 79.3% of interactions, compared with 17.2% for Outcall and 3.5% for Email. This concentration makes inbound operations the highest-impact area for workflow and staffing improvements.

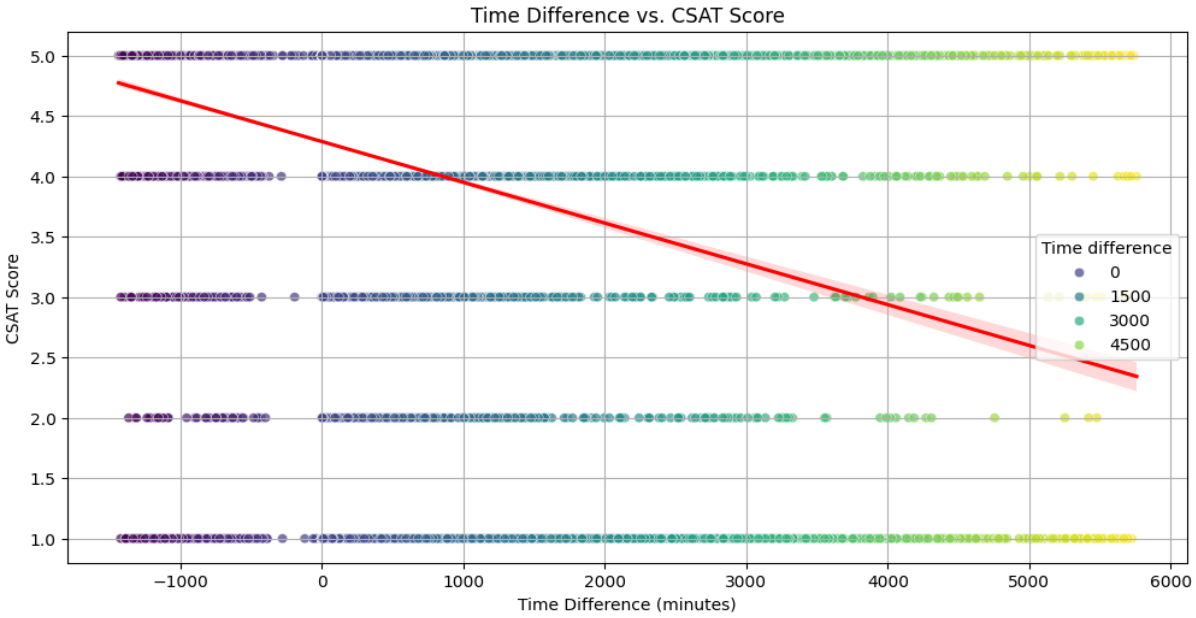



Business implication: review inbound queue times, repeat contacts and escalation patterns first. Preserve access to Outcall and Email while sizing staffing and process improvements to actual demand.

# 2. Response time & agent experience

## Response time vs. CSAT

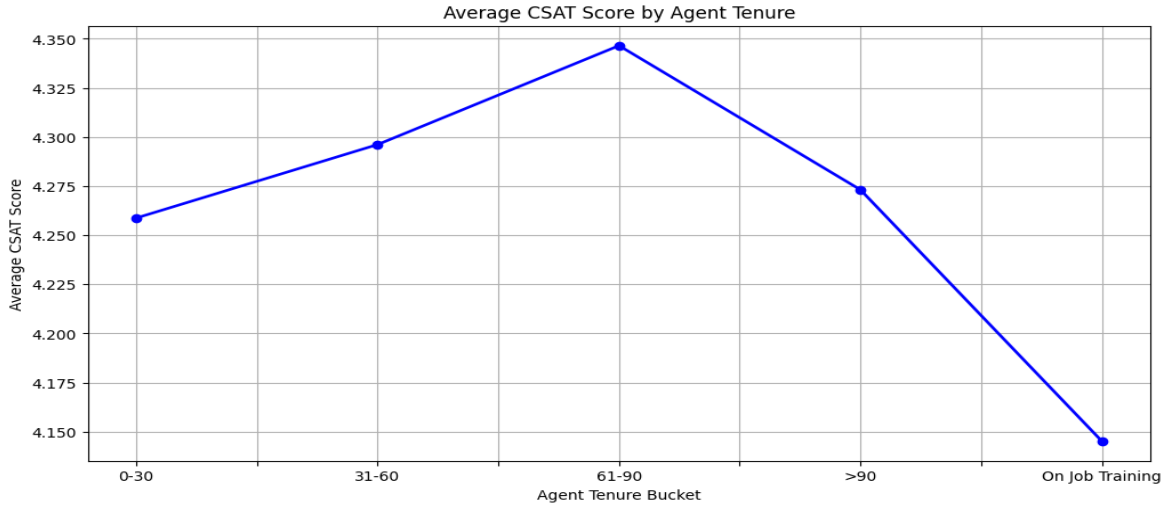
The scatter plot and fitted trend line slope downward. The notebook reports Spearman ρ = –0.1776 with p < 0.001, indicating a statistically significant negative association between time difference and CSAT. The effect is modest, so other factors also likely contribute to satisfaction.




Action: track response-time percentiles and extreme delays by shift; validate unusually large time differences before using them in KPIs. Pair speed metrics with resolution quality so that faster responses do not come at the expense of accurate support.

## Agent tenure pattern

Average CSAT rises through the 61–90 tenure bucket, then falls in the 90+ bucket; the On Job Training group is the lowest in the displayed trend. This suggests an opportunity to strengthen onboarding and investigate whether experienced-agent practices are being applied consistently.




Action: give new and training agents structured coaching, and review the practices of stronger tenure groups. Treat tenure as a signal for investigation, not as a standalone explanation for performance.

# 3. Statistical evidence & practical actions

## Hypothesis-test results reported in the notebook

| Relationship tested | Test result | Interpretation |
|---|---|---|
| Time difference vs. CSAT | Spearman ρ = –0.1776; p < 0.001 | Significant, modest negative association. |
| Agent shift vs. CSAT | Kruskal–Wallis H = 151.58; p = 9.34 × 10■[32] | CSAT distributions differ across shifts. |
| Item price vs. CSAT | Spearman ρ = –0.03395; p = 2.46 × 10■[23] | Statistically significant but extremely weak association; practical effect may be limited. |

## Recommended action plan

**1. Reduce avoidable delays.** Monitor response-time percentiles and extreme-delay cases by shift; investigate bottlenecks and data anomalies.

**2. Prioritize inbound operations.** Review staffing against inbound demand, queue length and peak periods because inbound represents about four-fifths of interactions.

**3. Target low-CSAT patterns.** Break down low ratings by channel, product category, shift and manager to find recurring issues before selecting interventions.

**4. Strengthen onboarding.** Provide structured coaching for On Job Training and early-tenure agents; share practices from stronger-performing groups.

**5. Use manager comparisons constructively.** Benchmark team-level patterns to identify coaching needs and repeatable practices, while accounting for workload and case mix.

## Data quality & scope notes

The supplied presentation reports 85,907 rows, 19 columns and no duplicate rows. It also identifies substantial missingness in order_id (18,232), order_date_time (68,693) and Product_category (68,711). These gaps may limit some segment-level interpretations. Validate missing-value handling and extreme time-difference values before operationalizing the findings.

The notebook contains populated EDA and three hypothesis tests. The supplied presentation notes that the ML model sections remain templated / do not provide completed model results; therefore, this report makes no claims about predictive model accuracy.

# Conclusion

**The clearest improvement opportunity is operational consistency: protect the strong CSAT base, focus first on inbound demand, reduce avoidable response delays, and use low ratings to target coaching and process fixes. Statistical significance should be considered alongside effect size, missing data and operational context.**